

recent stinging losses, which is the worst time to dis-invest because the low prices mean expected returns are high. Retail investors are especially prone to invest *pro-cyclically*, instead of *counter-cyclically*. One reason Vanguard has yet to introduce pure momentum factor funds is that small investors are likely to lose their shirts in poor factor timing.²⁶ George “Gus” Sauter, the former CIO of Vanguard says, “We have data that show that with narrowly defined investments, investors are not invested in them on the way up. They pile in at the top and then ride them on the way down.” An initial allocation to factor strategies is not enough; you must have the conviction to rebalance.

CAPM

The CAPM is a good place to show how the process works.²⁷ According to the CAPM, investors should hold the market portfolio with risk-free T-bills, as shown in Figure 14.6. Tracing out the combination of T-bills, which is the intercept on the y-axis, with the market portfolio, shown in the circle, yields the capital allocation line (CAL). Investors hold portfolios on this line. There is a special investor who holds a 100% market portfolio (Mr. Market). If you are more risk-averse than the market, you will hold less than 100% of the market and lie to the left of the market

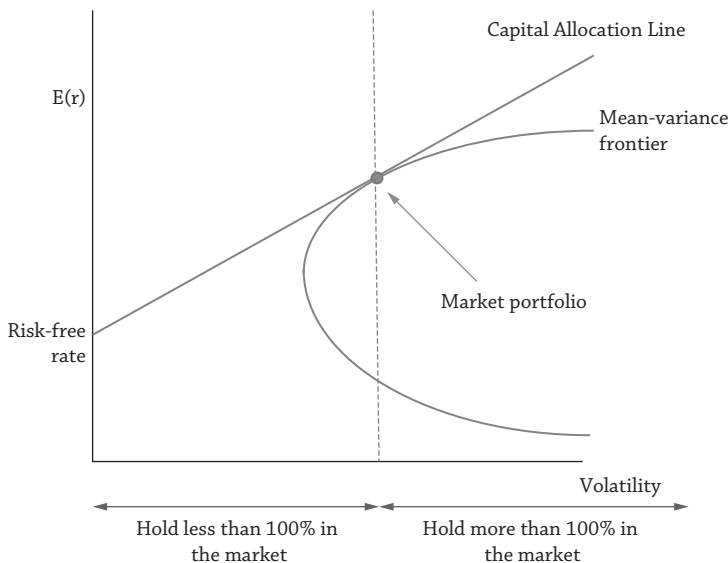


Figure 14.6

²⁶ Personal conversation with Sauter at the 2012 World Investment Forum. The quote is from Clark, K., “Listen Up, Bond Investors: Vanguard Investment Chief Gus Sauter Sees Low Returns Ahead for Some of His Most Popular Products,” CNNmoney.com, December 2012.

²⁷ See Chapter 6 for an exposition of the CAPM.